

SAE 3.03 – Concevoir un réseau informatique multi-sites

Rapport Technique – Partie FAI

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Configuration Réseau Multi-Sites FAI

Projet : Infrastructure MPLS VPN Multi-AS

Client : ABC Conseil & UC Exchange

Date : 8 Février 2026

Version : 1.0

1. Présentation du Projet

1.1 Objectifs

L'objectif de ce projet est de mettre en place une infrastructure réseau permettant l'interconnexion de sites distants pour deux clients distincts : - **ABC Conseil** : 3 sites à interconnecter - **UC Exchange** : 2 sites (Strasbourg et Metz)

1.2 Contraintes Techniques

- Isolation stricte entre les clients via VRF
- Utilisation de MPLS VPN pour le transport
- Interconnexion de 3 systèmes autonomes (AS 10, AS 20, AS 30)
- Protocoles de routage : RIP (AS 10, AS 20) et OSPF (AS 30)

1.3 Topologie Générale

Topologie générale du réseau

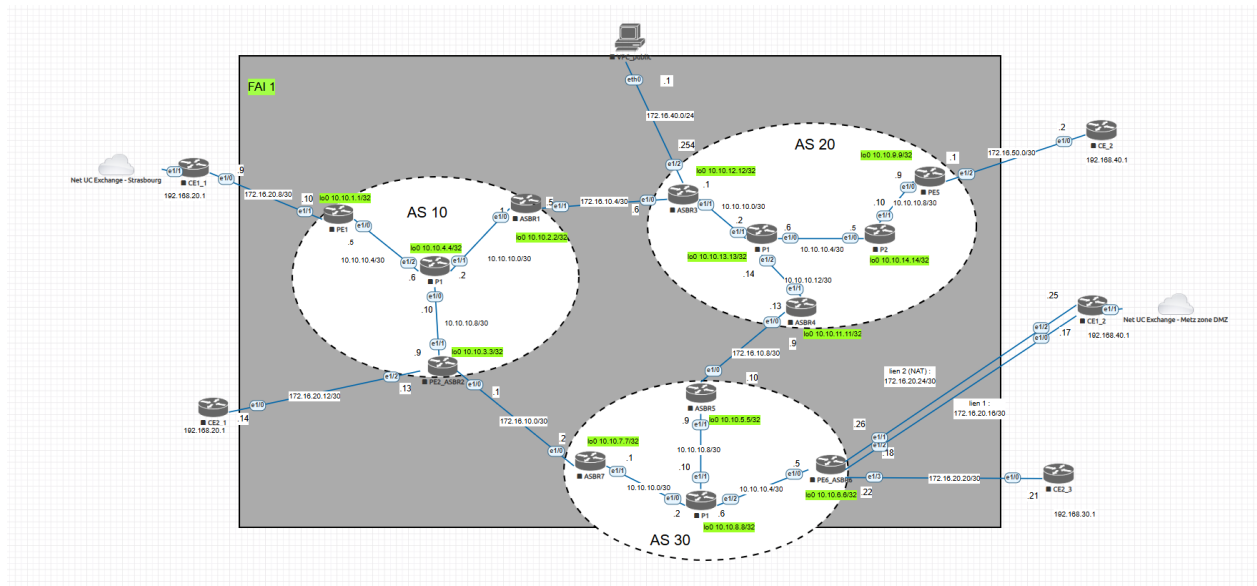


Figure 1 : Vue d'ensemble de l'infrastructure réseau

2. Architecture Réseau

2.1 Description des Systèmes Autonomes

AS 10 (AS 100)

- **Protocole IGP** : RIP version 2
- **Équipements** : PE1, P1 (Route Reflector), ASBR1, PE2/ASBR2

AS 20 (AS 200)

- **Protocole IGP** : RIP version 2
- **Équipements** : P1 (Route Reflector), P2, PE5, ASBR3, ASBR4

AS 30 (AS 300)

- **Protocole IGP** : OSPF
- **Équipements** : P1 (Route Reflector), PE6/ASBR6, ASBR5, ASBR7

Schéma des AS

Schéma des AS

Figure 2 : Détail des trois systèmes autonomes

3. Configuration des Services

3.1 Configuration MPLS

MPLS est activé sur toutes les interfaces de cœur de réseau pour permettre le transport des VPN.

```
interface EthernetX/X
 ip address X.X.X.X 255.255.255.252
 mpls ip
```

Configuration MPLS

```
!
interface Ethernet1/0
 ip address 10.10.10.10 255.255.255.252
 duplex full
 mpls ip
```

Figure 3 : Activation de MPLS sur les interfaces

3.2 Configuration BGP et Route Reflectors

Chaque AS dispose d'un Route Reflector pour optimiser la scalabilité du réseau BGP.

Configuration type d'un Route Reflector :

```
router bgp [AS_NUMBER]
 neighbor [PE_IP] remote-as [AS_NUMBER]
 neighbor [PE_IP] update-source Loopback0
 neighbor [PE_IP] route-reflector-client
 !
 address-family vpnv4
  neighbor [PE_IP] activate
  neighbor [PE_IP] send-community extended
  neighbor [PE_IP] route-reflector-client
 exit-address-family
```

Configuration Route Reflector

```
!
router bgp 100
 bgp log-neighbor-changes
 neighbor 10.10.1.1 remote-as 100
 neighbor 10.10.1.1 update-source Loopback0
 neighbor 10.10.1.1 route-reflector-client
 neighbor 10.10.2.2 remote-as 100
 neighbor 10.10.2.2 update-source Loopback0
 neighbor 10.10.2.2 route-reflector-client
 neighbor 10.10.3.3 remote-as 100
 neighbor 10.10.3.3 update-source Loopback0
 neighbor 10.10.3.3 route-reflector-client
 !
```

Figure 4 : Configuration du Route Reflector

Configuration sur les ASBR :

```
router bgp [AS_NUMBER]
 address-family vpnv4
```

```
neighbor [RR_IP] next-hop-self
exit-address-family
```

Configuration ASBR

```
address-family vpnv4
neighbor 10.10.4.4 activate
neighbor 10.10.4.4 send-community extended
neighbor 10.10.4.4 next-hop-self
neighbor 172.16.10.6 activate
neighbor 172.16.10.6 send-community both
exit-address-family
```

Figure 5 : Configuration des ASBR avec Next-Hop-Self

3.3 Configuration des VRF

VRF ABC_Conseil

Route Distinguisher : 10:1

Route Target : import/export 10:1

```
ip vrf ABC_conseil
rd 10:1
route-target export 10:1
route-target import 10:1
```

Sites connectés : - Site 1 via PE2 (AS 10) - Site 2 via PE5 (AS 20) - Site 3 via PE6 (AS 30)

VRF ABC Conseil

```
ip vrf ABC_conseil
rd 10:1
route-target export 10:1
route-target import 10:1
!
```

Figure 6 : Topologie VRF ABC_Conseil

VRF UC_Exchange

Route Distinguisher : 10:2

Route Target : import/export 10:2

```
ip vrf UC_Exchange
rd 10:2
route-target export 10:2
route-target import 10:2
```

Sites connectés : - Site Strasbourg via PE1 (AS 10) : 192.168.20.0/24 - Site Metz via PE6 (AS 30) : 192.168.40.0/24

VRF UC Exchange

```
ip vrf ABC_conseil
rd 10:1
route-target export 10:1
route-target import 10:1
!
```

Figure 7 : Topologie VRF UC_Exchange

3.4 Routage et Redistribution

PE1 - UC Exchange Strasbourg

```
ip route vrf UC_Exchange 192.168.20.0 255.255.255.0 172.16.20.9
```

```
router bgp 100
address-family ipv4 vrf UC_Exchange
redistribute connected
redistribute static
exit-address-family
```

PE6 - UC Exchange Metz (Double-Homing)

Configuration avec redondance sur deux liens :

```
ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.16.20.25
ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.16.20.17
```

```
router bgp 300
address-family ipv4 vrf UC_Exchange
redistribute connected
redistribute static
exit-address-family
```

Configuration PE6

```

router bgp 300
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  neighbor 10.10.8.8 remote-as 300
  neighbor 10.10.8.8 update-source Loopback0
  !
  address-family ipv4
    neighbor 10.10.8.8 activate
  exit-address-family
  !
  address-family vpnv4
    neighbor 10.10.8.8 activate
    neighbor 10.10.8.8 send-community extended
    neighbor 10.10.8.8 next-hop-self
  exit-address-family
  !
  address-family ipv4 vrf ABC_conseil
    redistribute connected
    redistribute rip
  exit-address-family
  !
  address-family ipv4 vrf UC_Exchange
    redistribute connected
    redistribute static
  exit-address-family

```

```

ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.16.20.25
ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.16.20.17
ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.168.20.25
ip route vrf UC_Exchange 192.168.40.0 255.255.255.0 172.168.20.17
!

```

Figure 8 : Configuration PE6 avec redondance

PE2, PE5, PE6 - ABC Conseil

Redistribution entre RIP et BGP :

```

router bgp [AS]
  address-family ipv4 vrf ABC_conseil
    redistribute connected
    redistribute rip
  exit-address-family

router rip
  address-family ipv4 vrf ABC_conseil
    redistribute bgp [AS] metric transparent
    network 172.16.0.0
    version 2
    no auto-summary
  exit-address-family

```

Redistribution RIP-BGP

```

router rip
!
address-family ipv4 vrf UC_Exchange
 redistribute connected
 redistribute static
 redistribute bgp 300
 no auto-summary
exit-address-family
!
address-family ipv4 vrf ABC_conseil
 redistribute bgp 300 metric 1
 network 172.16.0.0
 no auto-summary
 version 2
exit-address-family

```

```

address-family ipv4 vrf ABC_conseil
 redistribute connected
 redistribute rip
exit-address-family
!
address-family ipv4 vrf UC_Exchange
 redistribute connected
 redistribute static
exit-address-family
!

```

Figure 9 : Redistribution entre RIP et BGP

4. Tests et Validation

4.1 Vérification BGP

```

show ip bgp summary
show ip bgp vpnv4 all summary

```

État BGP

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.10.1.1	4	100	7	6	1	0	0	00:04:15	0
10.10.2.2	4	100	9	14	1	0	0	00:04:30	0
10.10.3.3	4	100	0	0	1	0	0	never	Idle

AS10_P1#

Figure 10 : Vérification des sessions BGP

4.3 Tests de Connectivité

Test VRF ABC_Conseil

```

ping vrf ABC_conseil [IP_SITE_2]
ping vrf ABC_conseil [IP_SITE_3]

```

Test ABC Conseil


```

CE2_1#ping 192.168.30.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/82/88 ms
CE2_1#ping 192.168.40.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
CE2_1#

```

Figure 12 : Tests de connectivité ABC_Conseil

Test VRF UC_Exchange

```

ping vrf UC_Exchange 192.168.40.1
traceroute vrf UC_Exchange 192.168.40.1

```

Test UC Exchange

```

CE1_1#ping 192.168.40.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
CE1_1#

```

Figure 13 : Tests de connectivité UC_Exchange

5. Problèmes Rencontrés et Résolutions

5.1 Routes VPNv4 non propagées

Symptôme : Les PE ne recevaient pas les routes VPNv4 des autres AS

Cause : Absence de la commande next-hop-self sur les ASBR

Solution :

```

router bgp [AS]
  address-family vpnv4
    neighbor [RR_IP] next-hop-self
  exit-address-family

```

5.2 Route Reflector Client manquant

Symptôme : Certains PE ne recevaient pas les routes des autres PE

Cause : Configuration incomplète du Route Reflector

Solution :

```

router bgp [AS]
  address-family vpnv4

```

```
neighbor [PE_IP] route-reflector-client
exit-address-family
```

6. Conclusion

6.1 Bilan du Projet

Le projet de mise en place d'une infrastructure MPLS VPN multi-AS a été réalisé avec succès :

Connectivité complète entre tous les sites clients

Isolation stricte entre les VRFs (ABC_Conseil et UC_Exchange)

Redondance mise en place sur le site critique (PE6 double-homing)

Scalabilité assurée via l'architecture Route Reflector

Interopérabilité entre différents protocoles IGP (RIP et OSPF)

Annexe : Commandes de Vérification

Vérification BGP

```
show ip bgp summary
show ip bgp vpnv4 all summary
show ip bgp vpnv4 vrf [VRF_NAME]
```

Vérification MPLS

```
show mpls ldp neighbor
show mpls forwarding-table
show mpls interfaces
```

Vérification VRF

```
show ip vrf
show ip route vrf [VRF_NAME]
show ip protocols vrf [VRF_NAME]
```

Tests de Connectivité

```
ping vrf [VRF_NAME] [IP_DESTINATION]
traceroute vrf [VRF_NAME] [IP_DESTINATION]
```